

LLM Observability & Eval

Trace analytics + drift + hallucination detection

June 3, 2025

Headline Result

5M traces / 1.79B tokens; PSI flags injected drift

Prepared for: Sai Veda

Publishing account: Nikeshk834

Project Context

Trace analytics + quality monitoring for LLM fleets at scale - a self-contained, fully offline reference implementation. It generates a realistic multi-tenant LLM trace firehose, lands it as partitioned Parquet, and runs four production concerns over it: cost/latency analytics (DuckDB, out-of-core), embedding-drift detection (PSI/KL), a hallucination detector (heuristic + logistic), and a per-model eval scorecard - with generated, product-grade dashboards.

This PDF dossier is intentionally visual and interview-ready: it captures the business problem, the technical architecture, the validation evidence, and the committed repository artifacts in a format that reads like a delivered project brief.

Repository slug

04-llm-observability-eval

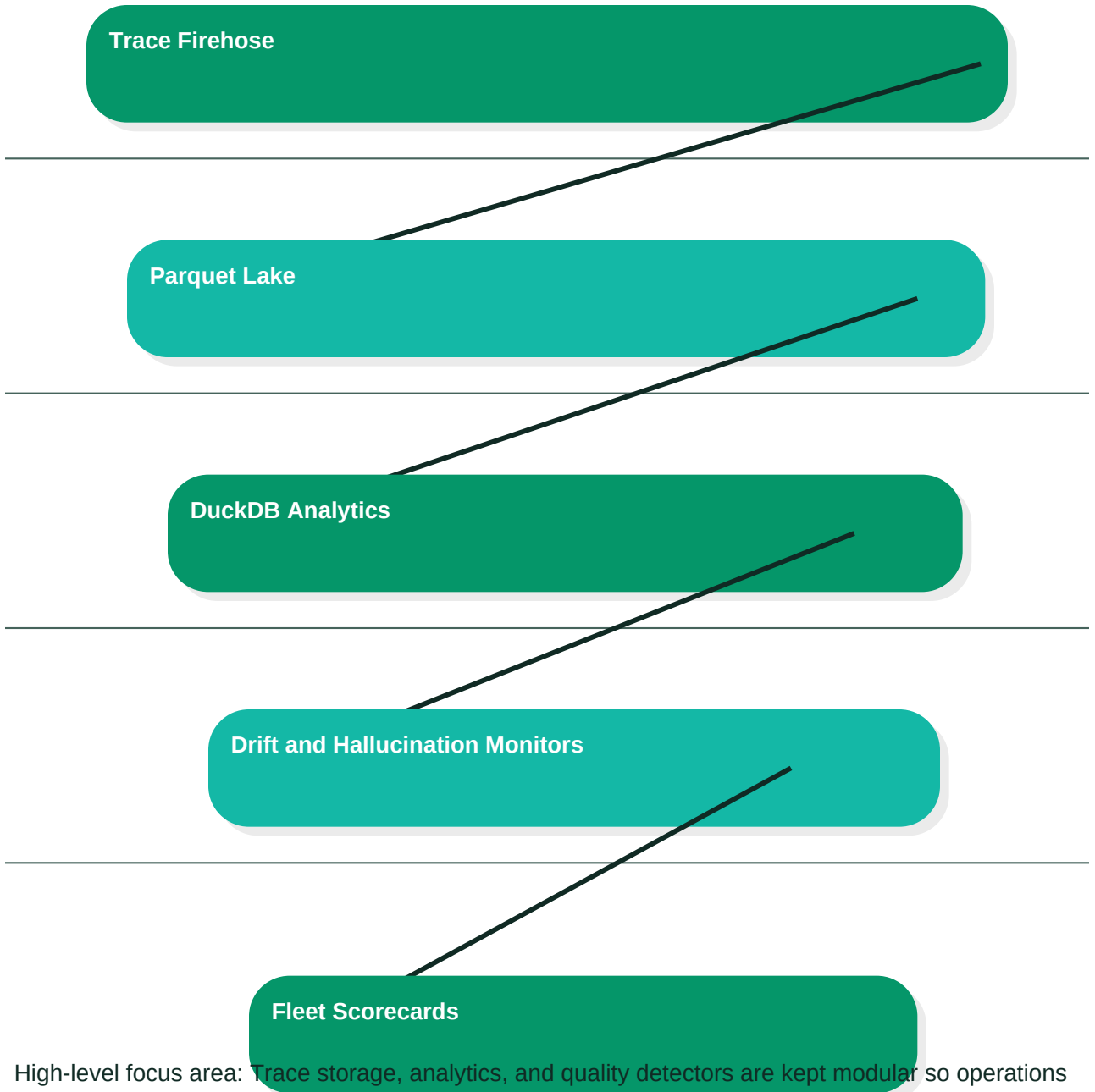
Validation index

21 tests

Architecture Storyboard

Trace storage, analytics, and quality detectors are kept modular so operations and model-risk teams can inspect the same pipeline.

Trace Firehose



Parquet Lake

DuckDB Analytics

Drift and Hallucination Monitors

Fleet Scorecards

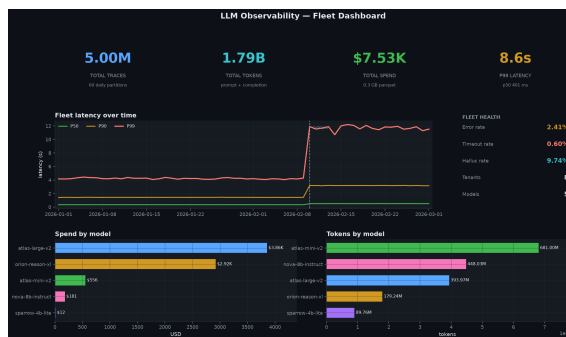
High-level focus area: Trace storage, analytics, and quality detectors are kept modular so operations and model-risk teams can inspect the same pipeline.

Validation And Visual Evidence

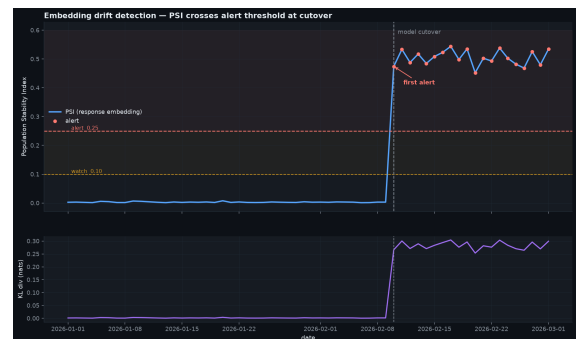
5M traces / 1.79B tokens; PSI flags injected drift

21 passing tests

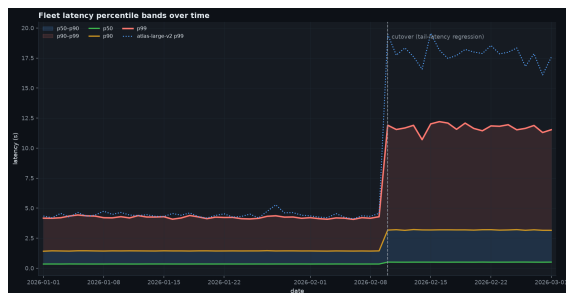
June 3, 2025



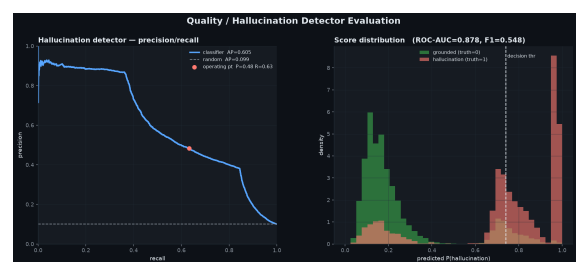
Fleet dashboard - KPI tiles, latency-over-time, spend/tokens by model



Drift detection - PSI crosses the alert threshold exactly at the model cutover



Latency percentile bands - the injected tail-latency regression is unmistakable



Hallucination detector - PR curve + score distribution vs a random baseline

Repository Handoff

Prepared for: Sal Veda

Publishing account: Nikeshk834

Project date: June 3, 2025

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Committed deliverables

- README.md - product overview, quickstart, and measured headline numbers
- ARCHITECTURE.md - detailed design and scale trade-offs
- tests/ - automated validation suite
- benchmarks/ - measured run artifacts and result tables
- assets/ - generated dashboards, charts, and screenshots
- PROJECT_DOCUMENT.pdf - styled project brief for sharing and review

This document is intentionally more visual than the repository Markdown. It is meant to read like a polished project brief: business framing first, architecture second, proof third, and delivery scope last.